Minimax Score Combination for Diffusion Sampling Toward Several Targets

author names withheld

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Abstract

Suppose we have m target distributions, know all of their true diffusion scores, and at every reverse-diffusion step use a convex combination of those scores. How should the weights be chosen so that the generated distribution is as close as possible to all targets simultaneously? We answer this question for the DDPM-type stochastic sampler. We prove an exact identity: the Kullback–Leibler divergence between the forward path of target i and the path of the sampler equals a term that does not depend on the weights plus $\frac{1}{2}\sum_t (1-\alpha_t)\, w_t^\top \Sigma_i^{(t)} w_t$, where $\Sigma_i^{(t)}$ is the Gram matrix of score differences under the noised target i. This yields a bound on the worst-case total-variation distance to the targets whose minimizer is the solution of a second-order cone program. The selected weights have an interpretation. At a pure weight, the relevant quantity is a Riemann sum for $\mathrm{KL}(q_1^i \| q_1^j) - \mathrm{KL}(q_T^i \| q_T^j)$, the Kullback–Leibler divergence between two noised targets, and for Gaussian targets with a common covariance, the mean of the combined distribution converges to the minimizer of the worst-case total variation. For two targets, the optimal constant weight is $\sqrt{\mathrm{KL}(p^1\|p^2)}/(\sqrt{\mathrm{KL}(p^1\|p^2)} + \sqrt{\mathrm{KL}(p^2\|p^1)})$ in the continuous-time limit. The proof is elementary and stays in discrete time.

Keywords: diffusion models, score combination, multi-objective sampling, minimax weights, Kullback–Leibler divergence, second-order cone programming

1. Introduction

Diffusion models generate data by running a reverse Markov chain driven by the scores of the noised data (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song and Ermon, 2019; Song et al., 2020). Several applications combine the scores of more than one model at sampling time, among them classifier and classifier-free guidance (Dhariwal and Nichol, 2021; Ho and Salimans, 2022), compositional generation (Liu et al., 2022), and the superposition of pretrained models (Skreta et al., 2025). These applications raise a basic question.

> *Suppose we have m target distributions, know all their true diffusion scores, and at every reverse-diffusion step use a convex combination of those scores. How should the weights be chosen so that the final generated distribution is as close as possible to all targets simultaneously?*

Existing combination rules are chosen by hand or tuned empirically, and existing convergence theory for diffusion samplers (Li et al., 2023; Chen et al., 2022, 2023a; Benton et al., 2023a) addresses a single target. Such theory can be applied to a combined score, but it is designed to control an estimation error. Weight selection is a different problem. In that theory, the score error enters through a uniform average over the T steps (Assumption 1 of Li et al. (2023)), and minimizing that average over the weights does not, in general, select the right weights (Section 4).

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We work with true scores, so that the only error in the output is the one created by the weights, and we study the DDPM-type (denoising diffusion probabilistic model) stochastic sampler analyzed by Li et al. (2023). Each step of this sampler is Gaussian with a mean that is linear in the score, while the reverse transition of each forward process has mean equal to the true-score step (Tweedie's formula). Replacing the score of target i by a combination therefore shifts the mean of a Gaussian at each step and changes nothing else, and the costs of these shifts add up along the path.

Contributions.

- We prove an exact decomposition (Theorem 1): for every weight schedule $\mathbf{w}$ and every target i, the path-space Kullback–Leibler divergence between forward process i and the sampler equals $\varepsilon_0^i + V_i(\mathbf{w})$, with ε_0^i independent of $\mathbf{w}$ and $V_i(\mathbf{w}) = \frac{1}{2}\sum_t (1-\alpha_t) w_t^\top \Sigma_i^{(t)} w_t$. Consequently, the worst-case total variation $\max_i \mathrm{TV}(q_1^i, p_1^{\mathbf{w}})$ is at most $\sqrt{\bar{\varepsilon}_0/2 + \max_i V_i(\mathbf{w})/2}$, a convex function of $\mathbf{w}$ whose minimizers are the minimax weights $\arg\min_{\mathbf{w}} \max_i V_i(\mathbf{w})$. Under the assumptions of Li et al. (2023), $\bar{\varepsilon}_0 \lesssim d^4 \log^6 T/T$. The proof follows the discrete-time approach of Li et al. (2023) and needs neither a Lipschitz condition on the scores nor any continuous-time tools.
- We interpret the minimax weights. At a pure weight e_j, $V_i(e_j)$ is a Riemann sum for $\mathrm{KL}(q_1^i \| q_1^j) - \mathrm{KL}(q_T^i \| q_T^j)$ (Lemma 2, via the de Bruijn identity). For Gaussian targets with a common covariance Σ_0, constant weights lose nothing, and the minimax weights place the mean of the combined distribution at the Chebyshev center of the target means in the metric Σ_0^{-1}. That center minimizes the worst-case total variation to the targets (Corollary 3).
- We show that the weights are computable. The minimax problem is a second-order cone program with $(T-1)m+1$ variables, its dual separates into $T-1$ independent m-dimensional quadratic programs, and replacing the Gram matrices by estimates costs at most twice the estimation error (Proposition 4). For $m = 2$, the minimax schedule is explicit up to a scalar root-finding problem, and the best constant weight has a closed form (Proposition 5).
- On Gaussian targets with an ill-conditioned covariance, the minimax weights attain the true optimum, while the weights obtained by minimizing the uniform time average are worse than most weights drawn at random (Section 4).

Related work. *Convergence theory.* Non-asymptotic analyses of diffusion samplers follow two routes. The first applies the Girsanov change of measure to the reverse stochastic differential equation (Chen et al., 2022, 2023a; Lee et al., 2022, 2023; Benton et al., 2023a; De Bortoli, 2022), and the second analyzes the discrete chain directly with elementary density-ratio arguments (Li et al., 2023, 2024). Deterministic samplers are treated by Chen et al. (2023b); Benton et al. (2023b) and Li et al. (2023), and the latter show that for deterministic samplers an L^2 score error alone cannot control total variation. Our proof follows the second route, and the identity of Theorem 1 is a discrete-time counterpart of the Girsanov formula for the relative entropy of two diffusions with a common diffusion coefficient.

Combining scores. Guidance forms an affine combination of a conditional and an unconditional score (Dhariwal and Nichol, 2021; Ho and Salimans, 2022). Wu et al. (2024) analyze it for Gaussian mixtures, and Bradley and Nakkiran (2024) interpret it as a predictor–corrector method. Liu et al. (2022) and Du et al. (2023) sum scores or energies to compose concepts, and Skreta et al. (2025)

reweight several models during sampling. These works propose combination rules, whereas we ask which weights are optimal for a stated distributional criterion.

Score matching and divergences. The link between score differences and relative entropy along the heat flow is the de Bruijn identity (Stam, 1959; Barron, 1986; Cover and Thomas, 2006), and Song et al. (2021) use it to relate weighted score matching to likelihood.

Multi-objective optimization. Selecting one point that serves several objectives is classical (Miettinen, 1998; Fliege and Svaiter, 2000; Désidéri, 2012) and has been developed in machine learning by Sener and Koltun (2018); Lin et al. (2019); Mahapatra and Rajan (2020); Navon et al. (2022). Our criterion is the minimax one. In the Gaussian case, it reduces to the minimum enclosing ball problem (Elzinga and Hearn, 1972), and in general to a second-order cone program (Lobo et al., 1998; Boyd, 2007).

2. Preliminaries

Notation. For probability measures P, Q, $\mathrm{TV}(P,Q) := \frac{1}{2}\int |dP - dQ|$ and $\mathrm{KL}(P\|Q) := \int \log \frac{dP}{dQ}\, dP \in [0,\infty]$. $\Delta^m := \{v \in \mathbb{R}^m_{\geq 0} : \sum_j v_j = 1\}$ is the probability simplex and e_j its j-th vertex. $\|A\|$ is the spectral norm, $\|x\|_M := (x^\top M x)^{1/2}$ for $M \succ 0$, and $a \lesssim b$ means $a \leq Cb$ for a universal constant C.

Forward processes. Let $p^1, \dots, p^m$ be distributions on $\mathbb{R}^d$. Each is diffused by the same forward chain. With $\beta_t \in (0,1)$, $\alpha_t := 1 - \beta_t$, and $\bar{\alpha}_t := \prod_{k \leq t} \alpha_k$,

$$X_0^i \sim p^i, \qquad X_t^i = \sqrt{\alpha_t}\, X_{t-1}^i + \sqrt{1-\alpha_t}\, W_t^i, \qquad W_t^i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d), \qquad 1 \leq t \leq T. \tag{1}$$

We write $q_t^i := \mathrm{law}(X_t^i)$ and $s_t^{i\star} := \nabla \log q_t^i$ for the true score. Recall that $X_t^i \overset{d}{=} \sqrt{\bar{\alpha}_t} X_0^i + \sqrt{1-\bar{\alpha}_t}\, Z$ with $Z \sim \mathcal{N}(0, I_d)$. We also use the inverse signal-to-noise ratio

$$\gamma_t := \frac{1-\bar{\alpha}_t}{\bar{\alpha}_t}, \qquad \text{so that} \qquad \gamma_t - \gamma_{t-1} = \frac{1-\alpha_t}{\bar{\alpha}_t} \quad (t \geq 2). \tag{2}$$

Combined score and sampler. A *weight schedule* is $\mathbf{w} = (w_2, \dots, w_T) \in (\Delta^m)^{T-1}$. It defines the combined score

$$s_{w_t}(x,t) := \sum_{j=1}^m [w_t]_j\, s_t^{j\star}(x), \tag{3}$$

which drives the DDPM-type sampler of Li et al. (2023, eq. (30)):

$$\begin{aligned} Y_T \sim \mathcal{N}(0, I_d), \qquad Y_{t-1} &= \frac{1}{\sqrt{\alpha_t}}\Big(Y_t + (1-\alpha_t)\, s_{w_t}(Y_t, t)\Big) + \sigma_t Z_t, \\ \sigma_t^2 &= \frac{1-\alpha_t}{\alpha_t}, \qquad t = T, \dots, 2, \end{aligned} \tag{4}$$

with $Z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_d)$. We write $p_t^{\mathbf{w}} := \mathrm{law}(Y_t)$. We call $\mathbf{w}$ *constant* if $w_t \equiv w$.

Objective. As in Li et al. (2023), we compare the output Y_1 with X_1^i. Since β_1 is small under the schedule (11) below, q_1^i and p^i are nearly identical. The quantity we want to make small is the worst-case gap

$$F_T(\mathbf{w}) := \max_{i \in [m]} \mathrm{TV}\big(q_1^i,\, p_1^{\mathbf{w}}\big). \tag{5}$$

Score differences and Gram matrices. Let $D_{ij}(x,t) := s_t^{j\star}(x) - s_t^{i\star}(x)$. Since $\sum_j [w_t]_j = 1$,

$$s_{w_t}(x,t) - s_t^{i\star}(x) = \sum_{j=1}^{m} [w_t]_j \, D_{ij}(x,t), \tag{6}$$

a linear function of w_t. Define the Gram matrix $\Sigma_i^{(t)} \in \mathbb{R}^{m\times m}$ and its diagonal entries by

$$\big[\Sigma_i^{(t)}\big]_{jk} := \mathbb{E}_{X\sim q_t^i}\big[D_{ij}(X,t)^\top D_{ik}(X,t)\big], \qquad \sigma_{ij,t}^2 := \big[\Sigma_i^{(t)}\big]_{jj}. \tag{7}$$

Each $\Sigma_i^{(t)}$ is positive semidefinite, its i-th row and column vanish, and by (6)

$$\mathbb{E}_{X\sim q_t^i}\big\|s_{w_t}(X,t) - s_t^{i\star}(X)\big\|_2^2 = w_t^\top \Sigma_i^{(t)} w_t. \tag{8}$$

Weighted by $1-\alpha_t$ and summed over the steps, these quadratic forms define

$$V_i(\mathbf{w}) := \frac{1}{2}\sum_{t=2}^{T} (1-\alpha_t)\, w_t^\top \Sigma_i^{(t)} w_t. \tag{9}$$

Relative Fisher information. Let $u_s^i := p^i * \mathcal{N}(0, sI_d)$ be target i smoothed by Gaussian noise of variance s, and let

$$\mathcal{J}_{ij}(s) := \mathbb{E}_{X\sim u_s^i}\big\|\nabla \log u_s^i(X) - \nabla \log u_s^j(X)\big\|_2^2 \tag{10}$$

be the relative Fisher information of u_s^i with respect to u_s^j.

Assumptions.

Assumption 1 Each p^i is absolutely continuous with finite second moment, and every entry of every $\Sigma_i^{(t)}$, $2 \le t \le T$, is finite.

Assumption 1 holds when each p^i is supported in a ball of radius R, since then $\|s_t^{j\star}(x)\|_2 \le (\|x\|_2 + \sqrt{\bar\alpha_t}R)/(1-\bar\alpha_t)$, and it holds for Gaussian targets. Rates require the setting of Li et al. (2023).

Assumption 2 For some constant $c_R > 0$ and all i, $\mathbb{P}(\|X_0^i\|_2 \le T^{c_R}) = 1$. The schedule is, for constants $c_0, c_1 > 0$ chosen as in Li et al. (2023) (in particular with c_1 large enough relative to c_0 that their property (39d), $\bar\alpha_T \le T^{-c_2}$, holds),

$$\beta_1 = \frac{1}{T^{c_0}}, \qquad \beta_t = \frac{c_1 \log T}{T}\min\Big\{\beta_1\Big(1 + \frac{c_1\log T}{T}\Big)^t,\, 1\Big\} \quad (t \ge 2). \tag{11}$$

3. Main results

3.1. The weight-selection rule

Each step of (4) draws Y_{t-1} from a Gaussian whose mean $\alpha_t^{-1/2}\big(x + (1-\alpha_t)s_{w_t}(x,t)\big)$ is linear in the score. The reverse transition of forward process i, $X_{t-1}^i \mid X_t^i = x$, is not Gaussian, but by Tweedie's formula its mean is the same expression with s_{w_t} replaced by $s_t^{i\star}$. Using the combined score in place of the true score of target i therefore shifts the mean of an otherwise unchanged comparison, and a mean shift of a Gaussian costs $\|\text{shift}\|^2/(2\sigma_t^2) = \frac{1}{2}(1-\alpha_t)\|s_{w_t} - s_t^{i\star}\|^2$ in relative entropy. Summing over steps and averaging over X_t^i gives $V_i(\mathbf{w})$. To be close to all targets at once, we minimize the worst case.

Minimax score combination.

1. For $2 \le t \le T$ and $i \in [m]$, compute (or estimate) the Gram matrices $\Sigma_i^{(t)}$ of (7).
2. Solve the convex program

$$\mathbf{w}^\star \in \operatorname*{arg\,min}_{\mathbf{w} \in (\Delta^m)^{T-1}} \max_{i \in [m]} V_i(\mathbf{w}). \tag{12}$$

3. Run the sampler (4) with the combined score $s_{w_t^\star}$.

3.2. Main theorem

Let $\mathbb{P}^i$ be the joint law of $(X_1^i, \dots, X_T^i)$, $\mathbb{P}^{\mathbf{w}}$ the joint law of $(Y_1, \dots, Y_T)$, and $p_{X_{t-1}^i | X_t^i}(\cdot \mid x)$ the conditional law of X_{t-1}^i given $X_t^i = x$. Let $\mu_t^{i\star}(x) := \alpha_t^{-1/2}\big(x + (1-\alpha_t) s_t^{i\star}(x)\big)$ be the mean of one sampler step driven by the true score of target i.

Theorem 1 *Suppose Assumption 1 holds. For every weight schedule $\mathbf{w} \in (\Delta^m)^{T-1}$ and every $i \in [m]$:*

(a) (Exact decomposition.)

$$\mathrm{KL}\big(\mathbb{P}^i \,\big\|\, \mathbb{P}^{\mathbf{w}}\big) = \varepsilon_0^i + V_i(\mathbf{w}), \tag{13}$$

where

$$\varepsilon_0^i := \mathrm{KL}\big(q_T^i \,\|\, \mathcal{N}(0, I_d)\big) + \sum_{t=2}^{T} \mathbb{E}_{x \sim q_t^i}\Big[\mathrm{KL}\Big(p_{X_{t-1}^i | X_t^i}(\cdot \mid x) \,\Big\|\, \mathcal{N}\big(\mu_t^{i\star}(x), \sigma_t^2 I_d\big)\Big)\Big] \tag{14}$$

does not depend on $\mathbf{w}$.

(b) (Bound on the worst-case gap.) $\mathrm{KL}(q_1^i \| p_1^{\mathbf{w}}) \le \varepsilon_0^i + V_i(\mathbf{w})$, *and with* $\bar{\varepsilon}_0 := \max_i \varepsilon_0^i$,

$$F_T(\mathbf{w}) \;\le\; B_T(\mathbf{w}) := \sqrt{\tfrac{1}{2}\bar{\varepsilon}_0 + \tfrac{1}{2}\max_{i \in [m]} V_i(\mathbf{w})}. \tag{15}$$

(c) (Convexity.) *B_T is convex on $(\Delta^m)^{T-1}$, and the set of its minimizers is the set of minimax schedules (12).*

(d) (Rate.) *If in addition Assumption 2 holds and $T \ge C_2 d^4 \log^6 T$ for a large enough constant C_2, then $\bar{\varepsilon}_0 \le C_1 d^4 \log^6 T / T$, and hence*

$$F_T(\mathbf{w}) \;\le\; C_3 \frac{d^2 \log^3 T}{\sqrt{T}} + \sqrt{\tfrac{1}{2}\max_{i \in [m]} V_i(\mathbf{w})} \tag{16}$$

for constants $C_1, C_2, C_3 > 0$ depending only on c_R, c_0, c_1.

Proof idea. The proof (Appendix A) follows the layout of the proof of Theorem 1 of Li et al. (2023), combined with the decomposition they use for the DDPM-type sampler in the proof of their Theorem 3. The chain rule writes the path divergence as $\mathrm{KL}(q_T^i \| \mathcal{N}(0, I_d))$ plus a sum over steps of divergences between the reverse transition of forward process i and the Gaussian transition of the sampler. By Tweedie's formula, the reverse transition has mean $\mu_t^{i\star}(x)$. Since the sampler's Gaussian has the same covariance as the true-score step, the divergence splits into the $\mathbf{w}$-independent term in (14) and the mean shift $\frac{1}{2}(1-\alpha_t)\|s_{w_t} - s_t^{i\star}\|^2$. Data processing and Pinsker's inequality give (15), and part (d) bounds ε_0^i with the estimates of Li et al. (2023).

3.3. Discussion

Interpretation of V_i. By (13), $V_i(\mathbf{w}) = \mathrm{KL}(\mathbb{P}^i \| \mathbb{P}^{\mathbf{w}}) - \varepsilon_0^i$ is the excess path divergence of the combined sampler over the true-score sampler of target i (whenever $\varepsilon_0^i < \infty$), so (12) minimizes the worst excess over the targets. At a vertex $w_t \equiv e_j$, the sampler is the true-score sampler of target j, and $V_i(e_j) = \frac{1}{2}\sum_t (1-\alpha_t)\sigma_{ij,t}^2$. Lemma 2 identifies $V_i(e_j)$.

Lemma 2 *For all i, j and $t \geq 2$, $(1-\alpha_t)\,\sigma_{ij,t}^2 = (\gamma_t - \gamma_{t-1})\,\mathcal{J}_{ij}(\gamma_t)$. Suppose moreover that each of p^i and p^j has bounded support or is Gaussian. Then:*

(a) For all $s > 0$, $\frac{d}{ds}\,\mathrm{KL}(u_s^i \| u_s^j) = -\frac{1}{2}\mathcal{J}_{ij}(s)$.

(b) $\frac{1}{2}\int_{\gamma_1}^{\gamma_T} \mathcal{J}_{ij}(s)\,ds = \mathrm{KL}(q_1^i \| q_1^j) - \mathrm{KL}(q_T^i \| q_T^j)$.

(c) If $\mathcal{J}_{ij}$ is non-increasing on $[\gamma_1, \gamma_T]$, then $V_i(e_j) \leq \mathrm{KL}(q_1^i \| q_1^j) - \mathrm{KL}(q_T^i \| q_T^j) \leq \mathrm{KL}(p^i \| p^j)$.

Thus, $V_i(e_j) = \frac{1}{2}\sum_{t\geq 2}(\gamma_t - \gamma_{t-1})\mathcal{J}_{ij}(\gamma_t)$ is a right-endpoint Riemann sum for $\mathrm{KL}(q_1^i \| q_1^j) - \mathrm{KL}(q_T^i \| q_T^j)$, and this difference tends to $\mathrm{KL}(p^i \| p^j)$ as $\gamma_1 \to 0$ and $\gamma_T \to \infty$ (by lower semi-continuity and data processing, and since $\mathrm{KL}(q_T^i \| q_T^j) \to 0$). We use *continuous-time limit* for the limit of vanishing step sizes with $\gamma_1 \to 0$ and $\gamma_T \to \infty$, in which $V_i(e_j)$ approaches $\mathrm{KL}(p^i \| p^j)$ whenever the Riemann sums converge, and Corollary 3 verifies this convergence for Gaussian targets under (11). For a constant schedule, $V_i(w) = w^\top G_i w$ with $G_i := \frac{1}{2}\sum_t (1-\alpha_t)\Sigma_i^{(t)}$, whose diagonal entries $[G_i]_{jj} = V_i(e_j)$ are these Riemann sums. The coefficient $1-\alpha_t$ produced by the proof of Theorem 1 turns the sum over steps into an integral in the variable γ, and the next result rests on this integral form.

Corollary 3 (Gaussian targets with a common covariance) *Let $p^i = \mathcal{N}(\mu_i, \Sigma_0)$ with $\Sigma_0 \succ 0$, and let $\mathbf{w}$ be constant, $w_t \equiv w$. Then:*

(a) $s_w(\cdot, t)$ is the true score of the noised version of $\mathcal{N}(\mu_w, \Sigma_0)$, where $\mu_w := \sum_j w_j \mu_j$. Hence, (4) is the true-score sampler of $\mathcal{N}(\mu_w, \Sigma_0)$.

(b) $V_i(w) = \|\mu_w - \mu_i\|_{M_T}^2$ with $M_T := \frac{1}{2}\sum_{t=2}^{T}(\gamma_t - \gamma_{t-1})(\Sigma_0 + \gamma_t I_d)^{-2}$, and $M_T \preceq \frac{1}{2}\big[(\Sigma_0 + \gamma_1 I_d)^{-1} - (\Sigma_0 + \gamma_T I_d)^{-1}\big] \preceq \frac{1}{2}\Sigma_0^{-1}$. Under the schedule (11), $M_T \to \frac{1}{2}\Sigma_0^{-1}$ as $T \to \infty$.

(c) Under (11), for minimax constant weights $w_T^\star \in \arg\min_{w\in\Delta^m} \max_i V_i(w)$, $\mu_{w_T^\star}$ converges as $T \to \infty$ to the Chebyshev center $c^\star$ of $\{\mu_1, \ldots, \mu_m\}$ in the norm $\|\cdot\|_{\Sigma_0^{-1}}$, and every limit point $\bar{w}$ of $w_T^\star$ satisfies $\mu_{\bar{w}} = c^\star$ and minimizes $\max_i \mathrm{TV}\big(p^i, \mathcal{N}(\mu_w, \Sigma_0)\big)$ over $w \in \Delta^m$.

(d) Constant schedules lose nothing. For every T, if $\mathbf{w}^\star$ is a minimax schedule over $(\Delta^m)^{T-1}$, then $\mu_{w_t^\star}$ does not depend on t, and $\mathbf{w}^\star$ produces the same sampler as a minimax constant weight.

In words, when the targets differ only in their means, the combined sampler is the true-score sampler of $\mathcal{N}(\mu_w, \Sigma_0)$, whose idealized worst-case gap $\max_i \mathrm{TV}(p^i, \mathcal{N}(\mu_w, \Sigma_0))$ is an increasing function of $\max_i \|\mu_w - \mu_i\|_{\Sigma_0^{-1}}$, and the minimax weights recover that metric in the limit.

Comparison with prior bounds. Applying Theorem 3 of Li et al. (2023) to the estimate $s_t = s_{w_t}$ bounds $\mathrm{TV}(q_1^i, p_1^{\mathbf{w}})$ by $C\, d^2 \log^3 T/\sqrt{T} + C\sqrt{d}\, \varepsilon_{\mathrm{score},i}(\mathbf{w}) \log^2 T$ for a constant C, where $\varepsilon^2_{\mathrm{score},i}(\mathbf{w}) = \frac{1}{T}\sum_{t=1}^T w_t^\top \Sigma_i^{(t)} w_t$ is the uniform average of their Assumption 1 (with any $w_1 \in \Delta^m$, since our sampler does not use w_1). Theorem 1 improves this bound in two ways. First, since $1 - \alpha_t \le c_1 \log T/T$ under (11), $V_i(\mathbf{w}) \le \frac{c_1}{2} \log T \cdot \varepsilon^2_{\mathrm{score},i}(\mathbf{w})$, so (16) has a score term of order $\sqrt{\log T}\, \varepsilon_{\mathrm{score},i}$ with no dependence on d. The same argument applies verbatim to any score estimate and is consistent with the Girsanov-based bounds of Chen et al. (2023a); Benton et al. (2023a). Second, and more important for weight selection, the uniform average gives the same weight to every step, whereas (9) gives step t the weight $1 - \alpha_t$. Under (11), the two differ on the initial steps, where $1 - \alpha_t$ is small and the scores are most sensitive to the data. For the Gaussian targets of Corollary 3, the uniform average equals $\|\mu_w - \mu_i\|^2_{M_{\mathrm{avg}}}$ with $M_{\mathrm{avg}} := \frac{1}{T}\sum_t \bar\alpha_t^{-1}(\Sigma_0 + \gamma_t I_d)^{-2}$, and in general M_{avg} is not proportional to Σ_0^{-1}. In Section 4, its condition number is 466 while that of Σ_0 is 30, and the weights it selects are far from optimal.

Computing the weights. Problem (12) is a small convex program, and its inputs are expectations that can be estimated by simulation.

Proposition 4 *Let $\Phi(\mathbf{w}) := \max_i V_i(\mathbf{w})$ and $H_t(\lambda) := \frac{1}{2}(1 - \alpha_t)\sum_{i=1}^m \lambda_i \Sigma_i^{(t)}$ for $\lambda \in \Delta^m$.*

(a) (Cone program.) *Writing $\frac{1}{2}(1 - \alpha_t)\Sigma_i^{(t)} = L_{i,t} L_{i,t}^\top$, problem (12) is the second-order cone program $\min\{\tau : \|(L_{i,2}^\top w_2, \dots, L_{i,T}^\top w_T)\|_2 \le \tau \ (i \in [m]),\ \mathbf{w} \in (\Delta^m)^{T-1}\}$, with $(T - 1)m + 1$ variables and m cone constraints.*

(b) (Separation across steps.)

$$\min_{\mathbf{w}} \Phi(\mathbf{w}) = \max_{\lambda \in \Delta^m} \sum_{t=2}^T \min_{v \in \Delta^m} v^\top H_t(\lambda) v,$$

the maximum is attained, and for every maximizer $\lambda^\star$ and every minimax schedule $\mathbf{w}^\star$, $w_t^\star \in \arg\min_{v \in \Delta^m} v^\top H_t(\lambda^\star) v$ for every t. Moreover, $\lambda_i^\star > 0$ only if $V_i(\mathbf{w}^\star) = \Phi(\mathbf{w}^\star)$.

(c) (Estimated inputs.) *Let $\widehat\Sigma_i^{(t)} \succeq 0$ be estimates, let $\widehat\Phi$ be Φ with $\Sigma_i^{(t)}$ replaced by $\widehat\Sigma_i^{(t)}$, and let $\delta := \max_i \frac{1}{2}\sum_t (1 - \alpha_t)\|\widehat\Sigma_i^{(t)} - \Sigma_i^{(t)}\|$. Every minimizer $\widehat{\mathbf{w}}$ of $\widehat\Phi$ satisfies $\Phi(\widehat{\mathbf{w}}) \le \min_{\mathbf{w}} \Phi(\mathbf{w}) + 2\delta$, and therefore $F_T(\widehat{\mathbf{w}}) \le \sqrt{\frac{1}{2}\bar\varepsilon_0 + \frac{1}{2}\min_{\mathbf{w}} \Phi(\mathbf{w}) + \delta}$.*

The inputs are available whenever the scores are. For target i and step t, draw $x = \sqrt{\bar\alpha_t}\, x_0 + \sqrt{1 - \bar\alpha_t}\, z$ with $x_0 \sim p^i$ (a training example) and $z \sim \mathcal{N}(0, I_d)$, evaluate the m scores at x, and average the Gram matrices of the differences $s_t^{j\star}(x) - s_t^{i\star}(x)$. With N draws, this costs $m^2(T -$

$1)N$ score evaluations ($N(T-1)$ evaluations of each of the m models on samples from each of the m targets), and the average is automatically positive semidefinite. Proposition 4(c) turns the operator-norm error of these averages into a guarantee. If a constant schedule is preferred, the program reduces to $\min_{w\in\Delta^m} \max_i w^\top G_i w$ in m variables. By part (b), the dual is an m-dimensional concave maximization in which each evaluation solves $T-1$ independent quadratic programs over Δ^m. The condition in (b) is necessary, and when each $H_t(\lambda^\star)$ is positive definite on $\{v : \sum_j v_j = 0\}$, it has a unique solution and determines $\mathbf{w}^\star$ (otherwise one solves the cone program of (a)). In the Gaussian case of Corollary 3, the problem is the classical minimum enclosing ball problem for m points (Elzinga and Hearn, 1972).

The case $m = 2$. With two targets, the dual of Proposition 4(b) is one-dimensional and everything is explicit. Write $w_t = (\theta_t, 1-\theta_t)$ and

$$\begin{aligned} a_t &:= \tfrac{1}{2}(1-\alpha_t)\,\sigma^2_{12,t}, & b_t &:= \tfrac{1}{2}(1-\alpha_t)\,\sigma^2_{21,t}, \\ \kappa_{12} &:= \sum_{t=2}^{T} a_t = V_1(e_2), & \kappa_{21} &:= \sum_{t=2}^{T} b_t = V_2(e_1). \end{aligned} \tag{17}$$

Then, $V_1(\mathbf{w}) = \sum_t a_t(1-\theta_t)^2$ and $V_2(\mathbf{w}) = \sum_t b_t\theta_t^2$.

Proposition 5 *Let $m = 2$ and suppose $a_t, b_t > 0$ for all t. For $\lambda \in [0,1]$, put $\theta_t(\lambda) := \lambda a_t/(\lambda a_t + (1-\lambda)b_t)$.*

(a) The minimax schedule is unique and equals $\theta_t^\star = \theta_t(\lambda^\star)$, where $\lambda^\star \in (0,1)$ is the unique root of the strictly decreasing function $\lambda \mapsto \sum_t a_t(1-\theta_t(\lambda))^2 - \sum_t b_t\theta_t(\lambda)^2$.

(b) $\theta^\star$ is constant in t if and only if a_t/b_t does not depend on t.

(c) Among constant schedules, the unique minimizer is

$$\bar\theta = \frac{\sqrt{\kappa_{12}}}{\sqrt{\kappa_{12}}+\sqrt{\kappa_{21}}}, \qquad \textit{with value} \quad \frac{\kappa_{12}\kappa_{21}}{(\sqrt{\kappa_{12}}+\sqrt{\kappa_{21}})^2}. \tag{18}$$

By Lemma 2, κ_{12} and κ_{21} are Riemann sums for the directed divergences between the noised targets, so in the continuous-time limit, $\bar\theta$ becomes $\sqrt{\mathrm{KL}(p^1\|p^2)}/\big(\sqrt{\mathrm{KL}(p^1\|p^2)}+\sqrt{\mathrm{KL}(p^2\|p^1)}\big)$. More weight goes to the target that is more expensive to miss. If the score of target 2 is a poor substitute for that of target 1 ($\mathrm{KL}(p^1\|p^2)$ large), target 1 gets more weight. For Gaussians with a common covariance, $a_t = b_t$, so $\theta_t^\star \equiv \bar\theta = \frac{1}{2}$, and asymmetric weights require asymmetric divergences, for example Gaussians with different covariances. The full schedule of part (a) costs one scalar root-finding problem (for example, bisection on λ), and it improves on $\bar\theta$ if and only if the relative cost a_t/b_t of the two targets changes along the chain.

4. Experiments

We test the weight-selection rule (12) on the discretized sampler (4). We compare the worst-case gap of the minimax weights with that of weights drawn uniformly at random from the simplex, of simple alternatives, and of the best weights found by direct search. Further details are in Appendix F.

4.1. Setting

Targets. All targets are Gaussian, $p^i = \mathcal{N}(\mu_i, S_i)$ on $\mathbb{R}^2$. We choose Gaussian targets because they are the one family in which the objective of the paper can be measured without error. Every score of a noised Gaussian is affine, so the combined score (3) is affine and the sampler maps Gaussians to Gaussians. We propagate the mean and covariance of Y_t through all T steps in closed form, and the Gram matrices (7) are also available in closed form. We take $d = 2$ because the total variation between two Gaussians in the plane can be computed by quadrature to four digits. The reported numbers are therefore the true worst-case gap $F_T(\mathbf{w})$ of (5) for the discretized sampler, with neither bounding nor sampling error, and a difference of 10^{-3} between two rules is a real difference.

Two families of instances. We use two families, chosen to sit on either side of what the theory proves.

- **Common covariance** ($S_i = \Sigma_0$ for all i). This is the setting of Corollary **??**. The experiment tests whether, at finite T, the minimax weights coincide with the best weights found by direct search over F_T.
- **Different covariances** (S_i drawn independently). Here Theorem 1 only says that the minimax weights minimize the bound B_T. The experiment measures how far the minimizer of B_T is from that of F_T. This is also the only family in which the weights of Proposition **??** can differ from $\frac{1}{2}$ and a schedule can differ from a constant weight (Corollary **??**(d), Proposition **??**(b)).

In both families the instance is random (Appendix **??**). The means are scaled so that the worst-case total variation of the minimax weights ranges from 0.05 to 0.99 across instances. The comparison therefore covers both targets that nearly coincide and targets that are far apart.

Number of targets. We take $m \in \{2, 3, 5\}$. Proposition 5 applies to the case $m = 2$. For a common covariance, this case is degenerate, since the minimax weight is $\frac{1}{2}$ by symmetry, and we include it as a check. The case $m = 3$ is the smallest for which the minimax weights of the common-covariance family differ from equal weights in general. The case $m = 5$ exceeds $d + 1 = 3$, the largest number of targets that can be active in a minimum enclosing ball in the plane, so some targets must receive weight zero.

A fixed instance. In addition, we keep one hand-built instance, designed to separate the weighting $1 - \alpha_t$ of V_i from the uniform time average discussed in Section 3.3. It has $m = 3$, $\Sigma_0 = \mathrm{diag}(30, 1)$, and means at Mahalanobis distance 2.86 from their center and 120° apart in the whitened coordinates, with target 1 on the high-variance axis, $\mu_i = 2.86\,\Sigma_0^{1/2}(\cos\phi_i, \sin\phi_i)$ with $\phi_i \in \{0°, 120°, 240°\}$. Three targets are needed, since for two targets the Chebyshev center is the midpoint in every metric.

Schedule. The noise schedule is (11) with $c_0 = 4$ and $c_1 = 8$, with $T = 1000$ for the random instances and $T = 3000$ for the fixed instance. With these constants, $\bar{\alpha}_T < 10^{-12}$, and the sampler driven by the score of a single target reproduces that target to within total variation 0.013 ($T = 1000$) and 0.005 ($T = 3000$). These numbers are the part of the error that no choice of weights can remove.

4.2. Rules compared

Every rule returns weights in the simplex and is evaluated by its true worst-case gap F_T.

- *Minimax schedule* (ours): the solution of (12) over $(\Delta^m)^{T-1}$, computed with the cone program of Proposition 4(a) from the exact Gram matrices.
- *Minimax constant* (ours): the solution of (12) restricted to constant schedules, $\min_{w\in\Delta^m} \max_i w^\top G_i w$.
- *Random*: w drawn uniformly from Δ^m and kept fixed for the whole run. Random weights are the natural reference for a rule that selects a point of the simplex, since they show the share of the simplex that is worse than the selected point.
- *Random, redrawn every step*: w_t drawn uniformly from Δ^m independently for each t. Over a thousand steps, these draws average out, and this rule behaves like equal weights, so we report it in the appendix only.
- *Equal*: $w = (1/m, \ldots, 1/m)$, the default in the absence of any information.
- *Uniform-average rule*: the constant weight minimizing $\max_i \frac{1}{T}\sum_{t=1}^{T} w^\top \Sigma_i^{(t)} w$, the quantity controlled by Assumption 1 of Li et al. (2023) (Section 3.3). It uses the same Gram matrices as our rule and differs only in the weighting of the steps.
- *Best constant weight found* (reference): a direct search over constant weights for the smallest F_T. It is not a practical rule, since it needs the output distribution of the sampler for every candidate weight. We use it as a reference for the smallest F_T attainable with a constant weight. It is an upper bound on $\min_{w\in\Delta^m} F_T(w)$, so the gap between our weights and this reference is a lower bound on our suboptimality among constant weights.

4.3. Protocol

Each of the six random-instance configurations (two families, $m \in \{2, 3, 5\}$) is run with 100 seeds, and given the seed every reported number is deterministic (Appendix **??**). The worst-case gap varies much more between instances than between rules, so we compare rules on the same instance. Table **??** reports means over instances, and Table **??** and Figure **??** report paired differences. We say that a weight beats another when its worst-case gap is smaller by more than 10^{-4}, a threshold above the quadrature error.

4.4. Results

Minimax weights against random weights. When the random rule is averaged over its draws, the minimax weights have a smaller worst-case gap than random weights on all but one of the 600 random instances (the exception has five targets with different covariances). The median paired difference is between 0.09 and 0.15 in total variation (Table 3). For a common covariance, none of the 30,000 individual random draws beats the minimax weights. For different covariances, between 4% and 6% of the draws do, and the best such draw is better than ours by 0.05 to 0.10. These draws show that outside Corollary 3 the minimax weights are not optimal.

Common covariance: the minimax weights are the best weights found. For a common covariance, the minimax weights agree with the reference to three digits in every row of Table 1, and the direct search improves on them on at most 8% of the instances, by at most 4×10^{-4}. This agreement is the finite-T form of Corollary 3(c). At $T = 1000$, the matrix M_T is close to Σ_0^{-1} but not proportional to it. The schedule and the constant weight coincide, as Corollary 3(d) states. Against equal

		Baselines			Minimax weights (ours)		Reference
Covariances	m	Random	Equal	Unif.-avg.	Constant	Schedule	Best found
Common	2	0.596	**0.457**	**0.457**	**0.457**	**0.457**	*0.457*
Common	3	0.755	0.682	0.652	**0.620**	**0.620**	*0.620*
Common	5	0.842	0.804	0.777	**0.739**	**0.739**	*0.739*
Different	2	0.665	0.544	0.581	0.545	**0.536**	*0.515*
Different	3	0.769	0.701	0.706	0.670	**0.660**	*0.634*
Different	5	0.833	0.794	0.780	0.750	**0.741**	*0.712*

Table 1: Worst-case total variation $F_T(\mathbf{w}) = \max_i \mathrm{TV}(q_1^i, p_1^{\mathbf{w}})$ of the sampler (lower is better), averaged over 100 random instances of m Gaussian targets in $d = 2$ with $T = 1000$. The best rule in each row is in bold. *Random*: a weight drawn uniformly from the simplex and kept fixed (mean over 100 draws per instance). *Equal*: $w = 1/m$. *Unif.-avg.*: the constant weight minimizing the uniform time average of the score error. *Minimax*: our weights (12), restricted to constant weights or as a full schedule. *Best found*: the constant weight with the smallest F_T found by direct search, shown as a reference, and a schedule can fall below it. All values are computed exactly (no sampling error), and paired differences are in Table 3.

weights, the gain is 0.06 for $m = 3$ and $m = 5$, and equal weights never beat ours. For $m = 2$, all rules other than random return $\frac{1}{2}$, as they must.

Different covariances: better than every baseline, short of the reference. For different covariances, the minimax schedule has the smallest mean gap among the rules, but the reference is lower by 0.02 to 0.03 on average, and the direct search finds a better constant weight on 93% to 98% of the instances. The median loss of our schedule with respect to the reference is 0.019, 0.024, and 0.024 for $m = 2, 3, 5$, comparable to our gain over equal weights for $m = 3$ and $m = 5$ (0.034 and 0.049 in median). Minimizing the bound B_T therefore recovers only part of the available gain over equal weights, and the shortfall is the experimental counterpart of the second limitation discussed in Section 5.

Two targets with different covariances. The configuration with two targets and different covariances is the least favorable. The constant weight $\bar{\theta}$ of Proposition 5(c) has the same mean gap as $\frac{1}{2}$ (0.545 against 0.544). The schedule of Proposition 5(a) improves on both, by 0.004 in median over equal weights, and equal weights beat it on 33% of the instances. For $m = 3$ and $m = 5$, equal weights beat the minimax schedule on 14% and 11% of the instances.

Schedule against constant weight. For different covariances, the minimax schedule improves on the minimax constant weight by 0.006 in median for each m. The benefit of a time-varying schedule in this family is therefore positive but small.

The uniform-average rule. On random instances, the uniform-average rule is worse than the minimax weights in every row with $m \geq 3$, by 0.03 to 0.05 on average, and for different covariances it is not better than equal weights for $m = 2$ and $m = 3$. The fixed instance (Table 2) isolates the reason. There, the minimax weights $[0.334, 0.333, 0.333]$ coincide with the best constant weight

Rule	Selected weights	Worst-case TV	Excess Mahalanobis distance
Best constant weight found	$[0.334, 0.333, 0.333]$	0.846	n/a
Minimax weights (12) (ours)	$[0.334, 0.333, 0.333]$	0.846	0.0%
Equal weights	$[0.333, 0.333, 0.333]$	0.846	0.0%
Uniform time average	$[0.000, 0.500, 0.500]$	0.968	$+49.9\%$
Uniform random w (mean)	n/a	0.945	$+37.6\%$

Table 2: The fixed instance of Section 4.1 ($m = 3$, $d = 2$, $\Sigma_0 = \mathrm{diag}(30, 1)$, $T = 3000$, $c_0 = 4$, $c_1 = 8$). The last column is the worst-case distance $\max_i \|\mu_w - \mu_i\|_{\Sigma_0^{-1}}$ relative to the best constant weight found. The last row averages over $10{,}000$ uniformly random weights (100 seeds with 100 draws each).

found and have a worst-case gap of 0.846. The uniform average induces a metric M_{avg} of condition number 466, against 30 for Σ_0, and selects $[0, \frac{1}{2}, \frac{1}{2}]$, assigning weight zero to the target on the high-variance axis. Its worst-case gap is 0.968, and its worst-case Mahalanobis distance to the targets is 50% larger than the optimum. Of $10{,}000$ weights drawn uniformly from the simplex, 79% have a smaller worst-case gap than the uniform-average rule, and none has a smaller one than the minimax weights. On this instance, equal weights are also optimal, by construction. The instance separates the two ways of weighting the steps. The comparison between the minimax weights and equal weights is in the rows $m = 3$ and $m = 5$ of Table 1.

Scope of the experiments. All targets are Gaussian and two-dimensional, and the scores are exact. The experiments do not test non-Gaussian targets, higher dimensions, other noise schedules or values of T, estimated Gram matrices (Proposition 4(c)), or learned scores.

5. Conclusion

For the DDPM-type sampler driven by a convex combination of m true scores, the path-space divergence to each target splits into a part that does not depend on the weights and the quadratic form $V_i(\mathbf{w}) = \frac{1}{2}\sum_t (1 - \alpha_t) w_t^\top \Sigma_i^{(t)} w_t$. Minimizing $\max_i V_i$ is a second-order cone program. It minimizes an upper bound on the worst-case total variation to the targets, its values at the vertices are Riemann sums for pairwise Kullback–Leibler divergences between the noised targets, and for Gaussian targets with a common covariance its minimizer places the mean of the combined distribution at the minimizer of the worst-case total variation in the limit. For two targets, the best constant weight is the ratio of square roots of directed divergences given in (18).

Limitations and future directions. The result concerns the stochastic sampler. For the probability-flow sampler, Li et al. (2023) show that an L^2 score error does not control total variation, so an analogue of Theorem 1 must also involve the Jacobians of the scores. Outside the Gaussian case, we minimize an upper bound, and bounding the excess true gap of the minimax weights for general targets is open. The term ε_0^i is small only when T is large, so for coarse schedules the bound (15) may be loose even though its minimizer is unaffected. Finally, we assume true scores. With learned scores, each constituent model adds its own estimation error, and that error need not be linear in the weights.

References

Andrew R Barron. Entropy and the central limit theorem. *The Annals of probability*, pages 336–342, 1986.

Joe Benton, Valentin De Bortoli, Arnaud Doucet, and George Deligiannidis. Nearly d-linear convergence bounds for diffusion models via stochastic localization. *arXiv preprint arXiv:2308.03686*, 2023a.

Joe Benton, George Deligiannidis, and Arnaud Doucet. Error bounds for flow matching methods. *arXiv preprint arXiv:2305.16860*, 2023b.

Stephen Boyd. Convex optimization of graph laplacian eigenvalues. In *Proceedings of the International Congress of Mathematicians Madrid, August 22–30, 2006*, pages 1311–1319. European Mathematical Society-EMS-Publishing House GmbH, 2007.

Arwen Bradley and Preetum Nakkiran. Classifier-free guidance is a predictor-corrector. *arXiv preprint arXiv:2408.09000*, 2024.

Hongrui Chen, Holden Lee, and Jianfeng Lu. Improved analysis of score-based generative modeling: User-friendly bounds under minimal smoothness assumptions. In *International Conference on Machine Learning*, pages 4735–4763. PMLR, 2023a.

Sitan Chen, Sinho Chewi, Jerry Li, Yuanzhi Li, Adil Salim, and Anru R Zhang. Sampling is as easy as learning the score: theory for diffusion models with minimal data assumptions. *arXiv preprint arXiv:2209.11215*, 2022.

Sitan Chen, Sinho Chewi, Holden Lee, Yuanzhi Li, Jianfeng Lu, and Adil Salim. The probability flow ode is provably fast. *Advances in Neural Information Processing Systems*, 36:68552–68575, 2023b.

Thomas M Cover and Joy A Thomas. Elements of information theory second edition solutions to problems. *Internet Access*, pages 19–20, 2006.

Valentin De Bortoli. Convergence of denoising diffusion models under the manifold hypothesis. *arXiv preprint arXiv:2208.05314*, 2022.

Jean-Antoine Désidéri. Multiple-gradient descent algorithm (mgda) for multiobjective optimization. *Comptes Rendus. Mathématique*, 350(5-6):313–318, 2012.

Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.

Yilun Du, Conor Durkan, Robin Strudel, Joshua B Tenenbaum, Sander Dieleman, Rob Fergus, Jascha Sohl-Dickstein, Arnaud Doucet, and Will Sussman Grathwohl. Reduce, reuse, recycle: Compositional generation with energy-based diffusion models and mcmc. In *International conference on machine learning*, pages 8489–8510. PMLR, 2023.

Bradley Efron. Tweedie's formula and selection bias. *Journal of the American Statistical Association*, 106(496):1602–1614, 2011.

D Jack Elzinga and Donald W Hearn. The minimum covering sphere problem. *Management science*, 19(1):96–104, 1972.

Jörg Fliege and Benar Fux Svaiter. Steepest descent methods for multicriteria optimization. *Mathematical methods of operations research*, 51(3):479–494, 2000.

Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.

Holden Lee, Jianfeng Lu, and Yixin Tan. Convergence for score-based generative modeling with polynomial complexity. *Advances in Neural Information Processing Systems*, 35:22870–22882, 2022.

Holden Lee, Jianfeng Lu, and Yixin Tan. Convergence of score-based generative modeling for general data distributions. In *International Conference on Algorithmic Learning Theory*, pages 946–985. PMLR, 2023.

Gen Li, Yuting Wei, Yuxin Chen, and Yuejie Chi. Towards faster non-asymptotic convergence for diffusion-based generative models. *arXiv preprint arXiv:2306.09251*, 2023.

Gen Li, Yu Huang, Timofey Efimov, Yuting Wei, Yuejie Chi, and Yuxin Chen. Accelerating convergence of score-based diffusion models, provably. *arXiv preprint arXiv:2403.03852*, 2024.

Xi Lin, Hui-Ling Zhen, Zhenhua Li, Qing-Fu Zhang, and Sam Kwong. Pareto multi-task learning. *Advances in neural information processing systems*, 32, 2019.

Shikun Liu, Stephen James, Andrew J Davison, and Edward Johns. Auto-lambda: Disentangling dynamic task relationships. *arXiv preprint arXiv:2202.03091*, 2022.

Miguel Sousa Lobo, Lieven Vandenberghe, Stephen Boyd, and Hervé Lebret. Applications of second-order cone programming. *Linear algebra and its applications*, 284(1-3):193–228, 1998.

Debabrata Mahapatra and Vaibhav Rajan. Multi-task learning with user preferences: Gradient descent with controlled ascent in pareto optimization. In *International Conference on Machine Learning*, pages 6597–6607. PMLR, 2020.

Kaisa Miettinen. No-preference methods. In *Nonlinear multiobjective optimization*, pages 67–76. Springer, 1998.

Aviv Navon, Aviv Shamsian, Idan Achituve, Haggai Maron, Kenji Kawaguchi, Gal Chechik, and Ethan Fetaya. Multi-task learning as a bargaining game. *arXiv preprint arXiv:2202.01017*, 2022.

Ozan Sener and Vladlen Koltun. Multi-task learning as multi-objective optimization. *Advances in neural information processing systems*, 31, 2018.

Maurice Sion. On general minimax theorems. 1958.

Marta Skreta, Lazar Atanackovic, Joey Bose, Alexander Tong, and Kirill Neklyudov. The superposition of diffusion models using the itô density estimator. In *International Conference on Learning Representations*, volume 2025, pages 67004–67057, 2025.

Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning*, pages 2256–2265. pmlr, 2015.

Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. *Advances in neural information processing systems*, 32, 2019.

Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*, 2020.

Yang Song, Conor Durkan, Iain Murray, and Stefano Ermon. Maximum likelihood training of score-based diffusion models. *Advances in neural information processing systems*, 34:1415–1428, 2021.

Aart J Stam. Some inequalities satisfied by the quantities of information of fisher and shannon. *Information and Control*, 2(2):101–112, 1959.

Yuchen Wu, Minshuo Chen, Zihao Li, Mengdi Wang, and Yuting Wei. Theoretical insights for diffusion guidance: A case study for gaussian mixture models. *arXiv preprint arXiv:2403.01639*, 2024.

Appendix A. Proof of Theorem 1

Following the layout of the proof of Theorem 1 of Li et al. (2023), and the decomposition used in their proof of Theorem 3, we work directly with the discrete-time chains and their transition kernels and never pass to a continuous-time limit. Throughout, fix a target $i \in [m]$ and a schedule $\mathbf{w} \in (\Delta^m)^{T-1}$, and abbreviate $X_t = X_t^i$, $q_t = q_t^i$, $s_t^\star = s_t^{i\star}$, and $\mu_t^\star = \mu_t^{i\star}$. All identities below are between elements of $[0, \infty]$. Under Assumption 1, every expected squared norm that appears is finite, so no expression of the form $\infty - \infty$ arises.

Preparation. Introduce the mappings

$$\mu_t^\star(x) = \frac{1}{\sqrt{\alpha_t}}\Big(x + (1-\alpha_t)s_t^\star(x)\Big), \qquad \mu_t^{\mathbf{w}}(x) := \frac{1}{\sqrt{\alpha_t}}\Big(x + (1-\alpha_t)s_{w_t}(x,t)\Big), \tag{19}$$

so that the update (4) reads $Y_{t-1} = \mu_t^{\mathbf{w}}(Y_t) + \sigma_t Z_t$. As in Li et al. (2023, eq. (75)), introduce the auxiliary vector

$$Y_{t-1}^\star := \mu_t^\star(Y_t) + \sigma_t Z_t, \tag{20}$$

the result of one sampler step from Y_t driven by the true score of target i. Conditionally on $Y_t = x$, both are Gaussian with the same covariance:

$$p_{Y_{t-1}|Y_t}(\cdot \mid x) = \mathcal{N}\big(\mu_t^{\mathbf{w}}(x), \sigma_t^2 I_d\big), \qquad p_{Y_{t-1}^\star|Y_t}(\cdot \mid x) = \mathcal{N}\big(\mu_t^\star(x), \sigma_t^2 I_d\big). \tag{21}$$

Finally, write $P_t(\cdot \mid x) := p_{X_{t-1}|X_t}(\cdot \mid x)$ for the reverse transition kernel of the forward chain.

Step 1: decomposition of the KL divergence. Both $X_1 \to \cdots \to X_T$ and $Y_T \to \cdots \to Y_1$ are Markov chains, and the time reversal of a Markov chain is again a Markov chain. The two joint laws therefore factorize as

$$p_{X_1,\dots,X_T}(x_1,\dots,x_T) = q_T(x_T)\prod_{t=2}^{T} P_t(x_{t-1} \mid x_t),$$
$$p_{Y_1,\dots,Y_T}(x_1,\dots,x_T) = \varphi(x_T)\prod_{t=2}^{T} p_{Y_{t-1}|Y_t}(x_{t-1} \mid x_t),$$

with φ the density of $\mathcal{N}(0, I_d)$, and the chain rule for the KL divergence gives the identity

$$\mathrm{KL}\big(\mathbb{P}^i \,\|\, \mathbb{P}^{\mathbf{w}}\big) = \mathrm{KL}\big(q_T \,\|\, \mathcal{N}(0, I_d)\big) + \sum_{t=2}^{T} \mathbb{E}_{x\sim q_t}\Big[\mathrm{KL}\big(P_t(\cdot \mid x) \,\big\|\, p_{Y_{t-1}|Y_t}(\cdot \mid x)\big)\Big]. \tag{22}$$

Identity (22) is the decomposition (79) of Li et al. (2023), except that we keep the identity at the level of paths instead of passing immediately to the marginals, and keeping the equality makes the dependence on $\mathbf{w}$ exact. It therefore suffices to understand, for each t, the dependence of the divergence between $P_t(\cdot \mid x)$ and the Gaussian kernel in (21) on the mean $\mu_t^{\mathbf{w}}(x)$ of that kernel.

Step 2: the mean of the reverse transition. The proof rests on the following lemma, a one-step form of Tweedie's formula (Efron, 2011). Its proof is in Appendix A.1.

Lemma 6 *For every $2 \le t \le T$ and every $x \in \mathbb{R}^d$,*

$$\mathbb{E}\big[X_{t-1} \mid X_t = x\big] = \int x'\, P_t(dx' \mid x) = \mu_t^\star(x). \tag{23}$$

In words, the true-score step $Y_{t-1}^\star$ has the correct conditional mean, and the discrepancy between $P_t(\cdot \mid x)$ and the Gaussian kernel lies only in the shape of $P_t(\cdot \mid x)$. That shape does not involve $\mathbf{w}$.

Step 3: quantifying the effect of the combined score. The next lemma is an elementary fact about Gaussian reference measures, proved in Appendix A.2.

Lemma 7 *Let P be a probability measure on $\mathbb{R}^d$ with finite first moment and mean m_P, and let $a, b \in \mathbb{R}^d$ and $\sigma > 0$. Then*

$$\mathrm{KL}\big(P \,\|\, \mathcal{N}(b, \sigma^2 I_d)\big) = \mathrm{KL}\big(P \,\|\, \mathcal{N}(a, \sigma^2 I_d)\big) + \frac{\|a-b\|_2^2 + 2\langle m_P - a,\, a - b\rangle}{2\sigma^2}, \tag{24}$$

where both sides may equal $+\infty$.

Apply Lemma 7 with $P = P_t(\cdot \mid x)$, $a = \mu_t^\star(x)$, $b = \mu_t^{\mathbf{w}}(x)$, and $\sigma = \sigma_t$. By Lemma 6, $m_P = a$, so the cross term vanishes and, in view of (21),

$$\mathrm{KL}\big(P_t(\cdot \mid x) \,\|\, p_{Y_{t-1}|Y_t}(\cdot \mid x)\big) = \mathrm{KL}\big(P_t(\cdot \mid x) \,\|\, p_{Y_{t-1}^\star|Y_t}(\cdot \mid x)\big) + \frac{\|\mu_t^\star(x) - \mu_t^{\mathbf{w}}(x)\|_2^2}{2\sigma_t^2}.$$

From (19), $\mu_t^\star(x) - \mu_t^{\mathbf{w}}(x) = \frac{1-\alpha_t}{\sqrt{\alpha_t}}\big(s_t^\star(x) - s_{w_t}(x,t)\big)$, and $\sigma_t^2 = (1-\alpha_t)/\alpha_t$, so the last term equals

$$\frac{(1-\alpha_t)^2/\alpha_t}{2(1-\alpha_t)/\alpha_t}\,\big\|s_{w_t}(x,t) - s_t^\star(x)\big\|_2^2 = \frac{1-\alpha_t}{2}\,\big\|s_{w_t}(x,t) - s_t^\star(x)\big\|_2^2.$$

Taking the expectation over $x \sim q_t$ and invoking (8), we arrive at

$$\begin{aligned}\mathbb{E}_{x\sim q_t}\Big[\mathrm{KL}\big(P_t(\cdot \mid x)\,\|\,p_{Y_{t-1}|Y_t}(\cdot \mid x)\big)\Big] &= \mathbb{E}_{x\sim q_t}\Big[\mathrm{KL}\big(P_t(\cdot \mid x)\,\|\,p_{Y^\star_{t-1}|Y_t}(\cdot \mid x)\big)\Big] \\ &\quad + \frac{1-\alpha_t}{2}\,w_t^\top \Sigma_i^{(t)} w_t.\end{aligned} \tag{25}$$

For a general score estimate s_t, Lemma 10 of Li et al. (2023) bounds the difference of the two expectations in (25) by a quantity of order $\exp(-c\,d\log T) + \frac{d\log^3 T}{T}\mathbb{E}_{q_t}\|s_t - s_t^\star\|_2^2$. For the Gaussian kernels of the DDPM-type sampler, the difference is available in closed form.

Step 4: proof of (a), (b), and (d). Substituting (25) into (22) and using (21) for $p_{Y^\star_{t-1}|Y_t}$ yields

$$\mathrm{KL}\big(\mathbb{P}^i \,\|\, \mathbb{P}^{\mathbf{w}}\big) = \varepsilon_0^i + \frac{1}{2}\sum_{t=2}^{T}(1-\alpha_t)\,w_t^\top \Sigma_i^{(t)} w_t = \varepsilon_0^i + V_i(\mathbf{w}),$$

with ε_0^i as in (14). The quantity ε_0^i involves only q_T, the kernels P_t, the maps $\mu_t^\star$, and the variances σ_t^2, none of which depends on $\mathbf{w}$. This proves (a).

For (b), the laws of Y_1 and X_1 are the images of $\mathbb{P}^{\mathbf{w}}$ and $\mathbb{P}^i$ under the projection onto the first coordinate, so the data-processing inequality gives $\mathrm{KL}(q_1^i \| p_1^{\mathbf{w}}) \le \mathrm{KL}(\mathbb{P}^i \| \mathbb{P}^{\mathbf{w}}) = \varepsilon_0^i + V_i(\mathbf{w})$. Pinsker's inequality $\mathrm{TV}(P,Q) \le \sqrt{\mathrm{KL}(P\|Q)/2}$ then gives

$$\mathrm{TV}\big(q_1^i, p_1^{\mathbf{w}}\big) \le \sqrt{\tfrac{1}{2}\varepsilon_0^i + \tfrac{1}{2}V_i(\mathbf{w})} \le \sqrt{\tfrac{1}{2}\bar{\varepsilon}_0 + \tfrac{1}{2}V_i(\mathbf{w})},$$

and taking the maximum over i gives (15).

For (d), ε_0^i is the right-hand side of (22) for the true-score sampler of target i, that is, for $w_t \equiv e_i$. Under Assumption 2 and $T \ge C_2 d^4 \log^6 T$, Lemma 3 of Li et al. (2023) gives $\mathrm{KL}(q_T \| \mathcal{N}(0, I_d)) \lesssim T^{-200}$, and inequality (92) in their proof of Theorem 3 gives, for each $2 \le t \le T$,

$$\mathbb{E}_{x\sim q_t}\Big[\mathrm{KL}\big(P_t(\cdot \mid x)\,\|\,p_{Y^\star_{t-1}|Y_t}(\cdot \mid x)\big)\Big] \lesssim \frac{d^4 \log^6 T}{T^2}.$$

Summing the $T-1$ terms gives $\varepsilon_0^i \le C_1 d^4 \log^6 T / T$ for every i. Inserting this into (15) and using $\sqrt{u+v} \le \sqrt{u} + \sqrt{v}$ yields (16), after adjusting C_1.

Step 5: convexity and the minimizers. Write $\frac{1}{2}(1-\alpha_t)\Sigma_i^{(t)} = L_{i,t}L_{i,t}^\top$ and let $z_i(\mathbf{w}) := (L_{i,2}^\top w_2, \dots, L_{i,T}^\top w_T)$, a linear function of $\mathbf{w}$. Then $V_i(\mathbf{w}) = \|z_i(\mathbf{w})\|_2^2$ and

$$B_T(\mathbf{w}) = \max_{i\in[m]}\Big\|\Big(\sqrt{\bar{\varepsilon}_0/2},\; z_i(\mathbf{w})/\sqrt{2}\Big)\Big\|_2,$$

a pointwise maximum of Euclidean norms of affine functions of $\mathbf{w}$, hence convex. Since $u \mapsto \sqrt{\bar{\varepsilon}_0/2 + u/2}$ is strictly increasing on $[0,\infty)$, B_T and $\max_i V_i$ have the same minimizers. This establishes (c) and completes the proof of Theorem 1. □

A.1. Proof of Lemma 6

Let $U := \sqrt{\alpha_t} X_{t-1}$, so that $X_t = U + \sqrt{1-\alpha_t}\, W_t$ with $W_t \sim \mathcal{N}(0, I_d)$ independent of U. Writing φ_v for the density of $\mathcal{N}(0, vI_d)$ and $v := 1 - \alpha_t$,

$$q_t(x) = \int \varphi_v(x-u)\, p_U(du) > 0.$$

Since $\nabla \varphi_v(y) = -\frac{y}{v}\varphi_v(y)$ is bounded, dominated convergence permits differentiation under the integral:

$$\nabla q_t(x) = -\frac{1}{v}\int (x-u)\, \varphi_v(x-u)\, p_U(du).$$

Dividing by $q_t(x)$ and recognizing $\varphi_v(x-u)\, p_U(du)/q_t(x)$ as the conditional law of U given $X_t = x$ (which has a finite mean because $\mathbb{E}\|U\|_2 < \infty$),

$$s_t^\star(x) = \nabla \log q_t(x) = -\frac{x - \mathbb{E}[U \mid X_t = x]}{1-\alpha_t}.$$

Rearranging, $\mathbb{E}[U \mid X_t = x] = x + (1-\alpha_t) s_t^\star(x)$, and dividing by $\sqrt{\alpha_t}$ gives (23). □

A.2. Proof of Lemma 7

If P is not absolutely continuous, both divergences are $+\infty$. Otherwise, for every $y \in \mathbb{R}^d$,

$$\log \frac{d\mathcal{N}(a, \sigma^2 I_d)}{d\mathcal{N}(b, \sigma^2 I_d)}(y) = \frac{\|y-b\|_2^2 - \|y-a\|_2^2}{2\sigma^2} = \frac{\|a-b\|_2^2 + 2\langle y-a,\, a-b\rangle}{2\sigma^2},$$

where we expanded $\|y-b\|_2^2 = \|(y-a)+(a-b)\|_2^2$. This function is affine in y, hence P-integrable with integral $(\|a-b\|_2^2 + 2\langle m_P - a, a-b\rangle)/(2\sigma^2)$. Writing $\log \frac{dP}{d\mathcal{N}(b,\sigma^2 I_d)} = \log \frac{dP}{d\mathcal{N}(a,\sigma^2 I_d)} + \log \frac{d\mathcal{N}(a,\sigma^2 I_d)}{d\mathcal{N}(b,\sigma^2 I_d)}$ and integrating against P gives (24). The negative part of the first logarithm is always P-integrable, so the identity holds in $(-\infty, \infty]$ and both sides are infinite together. □

Appendix B. Proof of Lemma 2

Part (a). Both u^i and u^j solve the heat equation $\partial_s u = \frac{1}{2}\Delta u$. Let $f := \log(u_s^i / u_s^j)$. Differentiating under the integral and using $\int \partial_s u_s^i = 0$,

$$\begin{aligned}\frac{d}{ds}\mathrm{KL}(u_s^i \| u_s^j) &= \frac{1}{2}\int f\, \Delta u_s^i - \frac{1}{2}\int \frac{u_s^i}{u_s^j}\, \Delta u_s^j \\ &= -\frac{1}{2}\int u_s^i\, \nabla \log u_s^i \cdot \nabla f + \frac{1}{2}\int u_s^i\, \nabla f \cdot \nabla \log u_s^j = -\frac{1}{2}\mathcal{J}_{ij}(s),\end{aligned}$$

where the middle equality integrates by parts and uses $\nabla(u_s^i/u_s^j) = (u_s^i/u_s^j)\nabla f$. The interchanges are justified as follows. If p^i, p^j are supported in a ball of radius R, then for s in a compact subset of $(0,\infty)$, the densities u_s^i, u_s^j and their first two derivatives are bounded above by $C(1+\|x\|_2^2)e^{-(\|x\|_2 - R)_+^2/(2s)}$, while $\|\nabla \log u_s^j(x)\|_2 \le (\|x\|_2 + R)/s$ and $|f(x)| \le C(1+\|x\|_2)$. All integrands are therefore dominated by integrable functions, and all boundary terms vanish. For Gaussian targets, the same holds by direct computation, and in the mixed case, $|f(x)| \le C(1+\|x\|_2^2)$, still dominated by the Gaussian tails of u_s^i. The same bounds show that $\mathcal{J}_{ij}$ is continuous on $(0,\infty)$.

The first identity and part (b). The first identity uses no assumption beyond the finiteness of $\sigma^2_{ij,t}$. Since $1-\bar\alpha_t = \bar\alpha_t\gamma_t$, we have $X^i_t \overset{d}{=} \sqrt{\bar\alpha_t}\big(X^i_0 + \sqrt{\gamma_t}Z\big)$, so q^i_t is the image of $u^i_{\gamma_t}$ under $x \mapsto \sqrt{\bar\alpha_t}x$: $q^i_t(y) = \bar\alpha_t^{-d/2}u^i_{\gamma_t}(y/\sqrt{\bar\alpha_t})$. Hence, $\nabla \log q^i_t(y) = \bar\alpha_t^{-1/2}(\nabla \log u^i_{\gamma_t})(y/\sqrt{\bar\alpha_t})$, and a change of variables gives $\sigma^2_{ij,t} = \bar\alpha_t^{-1}\mathcal{J}_{ij}(\gamma_t)$. Multiplying by $1-\alpha_t$ and using (2) gives $(1-\alpha_t)\sigma^2_{ij,t} = (\gamma_t - \gamma_{t-1})\mathcal{J}_{ij}(\gamma_t)$. The KL divergence is invariant under a common bijection, so $\mathrm{KL}(q^i_t\|q^j_t) = \mathrm{KL}(u^i_{\gamma_t}\|u^j_{\gamma_t})$. Since the derivative in (a) is continuous, (a) can be integrated over $[\gamma_1, \gamma_T]$, and this integration gives the stated integral identity.

Part (c). If $\mathcal{J}_{ij}$ is non-increasing, then $\mathcal{J}_{ij}(\gamma_t) \le \mathcal{J}_{ij}(s)$ for $s \in [\gamma_{t-1}, \gamma_t]$, so $(\gamma_t - \gamma_{t-1})\mathcal{J}_{ij}(\gamma_t) \le \int_{\gamma_{t-1}}^{\gamma_t} \mathcal{J}_{ij}(s)\,ds$. Summing over $2 \le t \le T$ and applying (b),

$$V_i(e_j) = \frac{1}{2}\sum_{t=2}^{T}(\gamma_t - \gamma_{t-1})\mathcal{J}_{ij}(\gamma_t) \le \mathrm{KL}(q^i_1\|q^j_1) - \mathrm{KL}(q^i_T\|q^j_T) \le \mathrm{KL}(q^i_1\|q^j_1) \le \mathrm{KL}(p^i\|p^j),$$

where the last step is the data-processing inequality for the common channel $x \mapsto \sqrt{\alpha_1}x + \sqrt{1-\alpha_1}W_1$. □

Appendix C. Proof of Corollary 3

Part (a). We have $q^i_t = \mathcal{N}(\sqrt{\bar\alpha_t}\mu_i, \Sigma_t)$ with $\Sigma_t := \bar\alpha_t\Sigma_0 + (1-\bar\alpha_t)I_d = \bar\alpha_t(\Sigma_0 + \gamma_t I_d)$, so $s^{i\star}_t(x) = -\Sigma_t^{-1}(x - \sqrt{\bar\alpha_t}\mu_i)$. Since $\sum_j w_j = 1$,

$$s_w(x,t) = \sum_j w_j\big(-\Sigma_t^{-1}(x - \sqrt{\bar\alpha_t}\mu_j)\big) = -\Sigma_t^{-1}\big(x - \sqrt{\bar\alpha_t}\mu_w\big),$$

the score of $\mathcal{N}(\sqrt{\bar\alpha_t}\mu_w, \Sigma_t)$, the law at step t of the forward process started from $\mathcal{N}(\mu_w, \Sigma_0)$.

Part (b). From the formula for the scores, $D_{ij}(x,t) = \sqrt{\bar\alpha_t}\,\Sigma_t^{-1}(\mu_j - \mu_i)$ does not depend on x, so by (6) and (8), $w^\top\Sigma_i^{(t)}w = \bar\alpha_t\,(\mu_w - \mu_i)^\top\Sigma_t^{-2}(\mu_w - \mu_i)$. Using $\Sigma_t^{-2} = \bar\alpha_t^{-2}(\Sigma_0 + \gamma_t I_d)^{-2}$ and (2),

$$(1-\alpha_t)\,\bar\alpha_t\,\Sigma_t^{-2} = \frac{1-\alpha_t}{\bar\alpha_t}(\Sigma_0 + \gamma_t I_d)^{-2} = (\gamma_t - \gamma_{t-1})(\Sigma_0 + \gamma_t I_d)^{-2},$$

which gives $V_i(w) = \|\mu_w - \mu_i\|^2_{M_T}$. All matrices involved are functions of Σ_0, so it suffices to compare eigenvalues. For an eigenvalue $\lambda > 0$ of Σ_0 and $t \ge 2$,

$$\int_{\gamma_{t-1}}^{\gamma_t}\frac{ds}{(\lambda+s)^2} = \frac{\gamma_t - \gamma_{t-1}}{(\lambda+\gamma_{t-1})(\lambda+\gamma_t)}, \qquad \text{so} \qquad \frac{\gamma_t - \gamma_{t-1}}{(\lambda+\gamma_t)^2} = \frac{\lambda+\gamma_{t-1}}{\lambda+\gamma_t}\int_{\gamma_{t-1}}^{\gamma_t}\frac{ds}{(\lambda+s)^2}.$$

The prefactor lies in $[\gamma_{t-1}/\gamma_t,\ 1]$. Summing over t and writing $\rho_T := \max_{2\le t\le T}\gamma_t/\gamma_{t-1}$,

$$\frac{1}{\rho_T}\Big[\frac{1}{\lambda+\gamma_1} - \frac{1}{\lambda+\gamma_T}\Big] \le \sum_{t=2}^{T}\frac{\gamma_t - \gamma_{t-1}}{(\lambda+\gamma_t)^2} \le \frac{1}{\lambda+\gamma_1} - \frac{1}{\lambda+\gamma_T}. \tag{26}$$

The right inequality is the stated upper bound on M_T. Under (11), $\gamma_1 = \beta_1/(1-\beta_1) \to 0$, $\gamma_T = \bar\alpha_T^{-1} - 1 \to \infty$ by property (39d) of Li et al. (2023), and

$$\frac{\gamma_t}{\gamma_{t-1}} = \frac{1}{\alpha_t}\cdot\frac{1-\bar\alpha_t}{1-\bar\alpha_{t-1}} \le \frac{1 + 4c_1\log T/T}{1 - c_1\log T/T} \to 1$$

by their properties (39a) and (39c). Hence, both sides of (26) tend to $1/\lambda$, that is, $M_T \to \frac{1}{2}\Sigma_0^{-1}$.

Part (c). Let $g_T(w) := \max_i \|\mu_w - \mu_i\|^2_{M_T}$ and $g(w) := \frac{1}{2}\max_i \|\mu_w - \mu_i\|^2_{\Sigma_0^{-1}}$. Since μ_w ranges over a bounded set and $M_T \to \frac{1}{2}\Sigma_0^{-1}$, $g_T \to g$ uniformly on Δ^m, so every limit point $\bar{w}$ of minimizers of g_T minimizes g. As w ranges over Δ^m, μ_w ranges over the convex hull of $\{\mu_i\}$, and this hull contains the Chebyshev center $c^\star$ (the unique minimizer over $\mathbb{R}^d$ of $\max_i \|c-\mu_i\|_{\Sigma_0^{-1}}$, by strict convexity of the squared norm). Hence, w minimizes g if and only if $\mu_w = c^\star$. Since Δ^m is compact and every limit point $\bar{w}$ of $w_T^\star$ has $\mu_{\bar{w}} = c^\star$, the whole sequence $\mu_{w_T^\star}$ converges to $c^\star$. Finally, for Gaussians with a common covariance, $\mathrm{TV}\big(\mathcal{N}(\mu,\Sigma_0),\mathcal{N}(\nu,\Sigma_0)\big) = 2\Phi_{\mathcal{N}}\big(\frac{1}{2}\|\mu-\nu\|_{\Sigma_0^{-1}}\big)-1$, where $\Phi_{\mathcal{N}}$ is the standard normal distribution function. The right-hand side is increasing in $\|\mu-\nu\|_{\Sigma_0^{-1}}$, so $\max_i \mathrm{TV}(p^i,\mathcal{N}(\mu_w,\Sigma_0))$ is an increasing function of $g(w)$ and has the same minimizers.

Part (d). By part (b) applied at each step, $V_i(\mathbf{w}) = \sum_{t=2}^T \|\mu_{w_t} - \mu_i\|^2_{A_t}$ with $A_t := \frac{1}{2}(\gamma_t - \gamma_{t-1})(\Sigma_0+\gamma_t I_d)^{-2} \succ 0$. Let $\lambda^\star$ be as in Proposition 4(b). For any norm $\|\cdot\|_A$, $\sum_i \lambda_i^\star \|x-\mu_i\|_A^2 = \|x-\mu_{\lambda^\star}\|_A^2 + \text{const}$, so $v^\top H_t(\lambda^\star)v = \sum_i \lambda_i^\star \|\mu_v - \mu_i\|^2_{A_t}$ is minimized over Δ^m if and only if $\mu_v = \mu_{\lambda^\star}$ (attained at $v=\lambda^\star$). By Proposition 4(b), every minimax schedule has $\mu_{w_t^\star} = \mu_{\lambda^\star}$ for all t. By part (a), the combined score at step t depends on w_t only through μ_{w_t}, so $\mathbf{w}^\star$ produces the same sampler, and the same values V_i, as the constant weight $\lambda^\star$. The constant weight $\lambda^\star$ is therefore a minimax constant weight. □

Appendix D. Proof of Proposition 4

Part (a). As in Step 5 of Appendix A, $V_i(\mathbf{w}) = \|z_i(\mathbf{w})\|_2^2$ with z_i linear, so $\min_{\mathbf{w}} \max_i V_i(\mathbf{w})$ equals the square of $\min\{\tau : \|z_i(\mathbf{w})\|_2 \le \tau\ (i\in[m]),\ \mathbf{w}\in(\Delta^m)^{T-1}\}$, and both problems have the same minimizers.

Part (b). Let $L(\mathbf{w},\lambda) := \sum_i \lambda_i V_i(\mathbf{w}) = \sum_{t=2}^T w_t^\top H_t(\lambda) w_t$, so that $\Phi(\mathbf{w}) = \max_{\lambda\in\Delta^m} L(\mathbf{w},\lambda)$. The function L is continuous, convex in $\mathbf{w}$ and linear in λ, and both $(\Delta^m)^{T-1}$ and Δ^m are compact and convex, so Sion's minimax theorem (Sion, 1958) gives $\min_{\mathbf{w}}\Phi(\mathbf{w}) = \max_\lambda \psi(\lambda)$ with $\psi(\lambda) := \min_{\mathbf{w}} L(\mathbf{w},\lambda) = \sum_t \min_{v\in\Delta^m} v^\top H_t(\lambda) v$, and the maximum is attained at some $\lambda^\star$ because ψ is continuous on a compact set. For any minimax schedule $\mathbf{w}^\star$,

$$\Phi(\mathbf{w}^\star) = \psi(\lambda^\star) = \min_{\mathbf{w}} L(\mathbf{w},\lambda^\star) \le L(\mathbf{w}^\star,\lambda^\star) \le \Phi(\mathbf{w}^\star),$$

so equality holds throughout. Equality in the first inequality shows $\mathbf{w}^\star \in \arg\min_{\mathbf{w}} L(\mathbf{w},\lambda^\star)$, and since $L(\cdot,\lambda^\star)$ is a sum of functions of the separate w_t, each $w_t^\star$ minimizes $v^\top H_t(\lambda^\star)v$ over Δ^m. Equality in the second shows $\sum_i \lambda_i^\star V_i(\mathbf{w}^\star) = \max_i V_i(\mathbf{w}^\star)$, so $\lambda_i^\star = 0$ whenever $V_i(\mathbf{w}^\star) < \Phi(\mathbf{w}^\star)$.

Part (c). For every $\mathbf{w}\in(\Delta^m)^{T-1}$ and i, since $\|w_t\|_2 \le 1$,

$$\big|\widehat{V}_i(\mathbf{w}) - V_i(\mathbf{w})\big| = \Big|\tfrac{1}{2}\sum_t (1-\alpha_t)\, w_t^\top\big(\widehat{\Sigma}_i^{(t)} - \Sigma_i^{(t)}\big)w_t\Big| \le \tfrac{1}{2}\sum_t (1-\alpha_t)\big\|\widehat{\Sigma}_i^{(t)} - \Sigma_i^{(t)}\big\| \le \delta,$$

so $|\widehat{\Phi}-\Phi| \le \delta$ on $(\Delta^m)^{T-1}$. For any minimax schedule $\mathbf{w}^\star$, $\Phi(\widehat{\mathbf{w}}) \le \widehat{\Phi}(\widehat{\mathbf{w}}) + \delta \le \widehat{\Phi}(\mathbf{w}^\star) + \delta \le \Phi(\mathbf{w}^\star) + 2\delta$. The bound on $F_T(\widehat{\mathbf{w}})$ follows from (15). □

Appendix E. Proof of Proposition 5

Write $A(\theta) := \sum_t a_t(1-\theta_t)^2 = V_1(\mathbf{w})$ and $B(\theta) := \sum_t b_t\theta_t^2 = V_2(\mathbf{w})$ for $\theta \in [0,1]^{T-1}$, and $L(\theta,\lambda) := \lambda A(\theta) + (1-\lambda)B(\theta)$.

Part (a). For each $\lambda \in [0,1]$, $L(\cdot,\lambda)$ is separable, with t-th term $\lambda a_t(1-\theta_t)^2 + (1-\lambda)b_t\theta_t^2$, a strictly convex quadratic because $\lambda a_t + (1-\lambda)b_t > 0$. Its unconstrained minimizer is $\theta_t(\lambda) \in [0,1]$, so $\theta(\lambda)$ is the unique minimizer of $L(\cdot,\lambda)$ over $[0,1]^{T-1}$. Each $\theta_t(\lambda) = \big(1 + \frac{(1-\lambda)b_t}{\lambda a_t}\big)^{-1}$ is strictly increasing in λ, with $\theta_t(0) = 0$ and $\theta_t(1) = 1$. Hence, $g(\lambda) := A(\theta(\lambda)) - B(\theta(\lambda))$ is continuous and strictly decreasing, with $g(0) = \sum_t a_t > 0$ and $g(1) = -\sum_t b_t < 0$, so it has a unique root $\lambda^\star \in (0,1)$.

At $\theta^\star := \theta(\lambda^\star)$, we have $A(\theta^\star) = B(\theta^\star)$, hence $\max\{A,B\}(\theta^\star) = L(\theta^\star,\lambda^\star) = \min_\theta L(\theta,\lambda^\star)$. For any θ, $\max\{A,B\}(\theta) \geq L(\theta,\lambda^\star) \geq L(\theta^\star,\lambda^\star)$, so $\theta^\star$ is a minimax schedule. If θ' is another one, then $L(\theta',\lambda^\star) \leq \max\{A,B\}(\theta') = \max\{A,B\}(\theta^\star) = \min_\theta L(\theta,\lambda^\star)$, so θ' minimizes $L(\cdot,\lambda^\star)$ and equals $\theta^\star$ by uniqueness.

Part (b). Since $\lambda^\star \in (0,1)$, $\theta_t^\star = \big(1 + \frac{1-\lambda^\star}{\lambda^\star} \cdot \frac{b_t}{a_t}\big)^{-1}$ is constant in t if and only if b_t/a_t is.

Part (c). For $\theta_t \equiv \theta \in [0,1]$, $\max\{A,B\} = \big(\max\{(1-\theta)\sqrt{\kappa_{12}}, \theta\sqrt{\kappa_{21}}\}\big)^2$, the square of the maximum of a strictly decreasing and a strictly increasing linear function of θ. It is uniquely minimized where they cross, at $(1-\bar\theta)\sqrt{\kappa_{12}} = \bar\theta\sqrt{\kappa_{21}}$, and solving for $\bar\theta$ gives (18). The value is $(\bar\theta\sqrt{\kappa_{21}})^2 = \kappa_{12}\kappa_{21}/(\sqrt{\kappa_{12}} + \sqrt{\kappa_{21}})^2$. □

Appendix F. Experimental details

This appendix gives the details needed to reproduce Section 4: the instances, the way each quantity is computed, the random seeds, the statistics, the paired results, and the computing environment.

F.1. Instances

Schedule. All runs use (11) with $c_0 = 4$ and $c_1 = 8$, so that $\beta_1 = T^{-4}$. For $T = 1000$, this gives $\beta_1 = 10^{-12}$, $\max_t(1-\alpha_t) = 0.055$, and $\bar\alpha_T = 3.4 \times 10^{-13}$, and for $T = 3000$, $\max_t(1-\alpha_t) = 0.021$ and $\bar\alpha_T = 4.4 \times 10^{-15}$.

Common covariance. Draw κ with $\log\kappa \sim \mathrm{Unif}[0, \log 30]$ and an angle $\vartheta \sim \mathrm{Unif}[0,\pi)$, and let $A := R_\vartheta\,\mathrm{diag}(\sqrt{\kappa}, 1)$, where R_ϑ is the rotation by ϑ. Then $\Sigma_0 := AA^\top$ and $\mu_i := 1.5\,A\,g_i$ with $g_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_2)$, $i \in [m]$. In the whitened coordinates, the means are thus $1.5\,g_i$, independently of Σ_0.

Different covariances. For each $i \in [m]$ independently, draw $\log\kappa_i \sim \mathrm{Unif}[0, \log 10]$, $\log s_i \sim \mathrm{Unif}[\log\frac{1}{2}, \log 2]$ and $\vartheta_i \sim \mathrm{Unif}[0,\pi)$, and set $S_i := s_i\,R_{\vartheta_i}\mathrm{diag}(\sqrt{\kappa_i}, 1/\sqrt{\kappa_i})R_{\vartheta_i}^\top$, a matrix with condition number κ_i and determinant s_i^2. The means are $\mu_i := 1.5\,g_i$ with $g_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_2)$.

Fixed instance. $m = 3$, $\Sigma_0 = \mathrm{diag}(30, 1)$, $\mu_i = 2.86\,\Sigma_0^{1/2}(\cos\phi_i, \sin\phi_i)$ with $\phi_i \in \{0°, 120°, 240°\}$, that is, $\mu_1 = (15.66, 0)$ and $\mu_{2,3} = (-7.83, \pm 2.48)$, and $T = 3000$. The radius 2.86 corresponds to an idealized worst-case total variation $2\Phi_{\mathcal{N}}(1.43) - 1 = 0.847$ at the

Chebyshev center. The orientation matters for the uniform-average rule. With target 1 on the high-variance axis, the metric M_{avg} shrinks the direction that separates target 1 from the other two, and the rule assigns weight zero to target 1.

Choice of ranges. Over the 100 instances of each configuration, the worst-case gap of the minimax weights ranges from 0.05 to 0.99, so the comparison covers targets that nearly coincide as well as targets that are far apart. Condition numbers up to 30 separate the metric Σ_0^{-1} of Corollary 3 from the Euclidean one.

F.2. Exact computation

Scores. For $p^j = \mathcal{N}(\mu_j, S_j)$, we have $q_t^j = \mathcal{N}(\sqrt{\bar\alpha_t}\mu_j, \Sigma_t^j)$ with $\Sigma_t^j := \bar\alpha_t S_j + (1-\bar\alpha_t) I_d$, and $s_t^{j\star}(x) = -P_t^j(x - \sqrt{\bar\alpha_t}\mu_j)$ with $P_t^j := (\Sigma_t^j)^{-1}$.

Output of the sampler. With these scores, one step of (4) is the affine map $Y_{t-1} = K_t Y_t + k_t + \sigma_t Z_t$, where

$$K_t := \frac{1}{\sqrt{\alpha_t}}\Big(I_d - (1-\alpha_t)\sum_j [w_t]_j P_t^j\Big), \qquad k_t := \frac{(1-\alpha_t)\sqrt{\bar\alpha_t}}{\sqrt{\alpha_t}}\sum_j [w_t]_j P_t^j \mu_j.$$

Hence, $Y_t \sim \mathcal{N}(m_t, C_t)$ for every t, with $m_T = 0$, $C_T = I_d$ and

$$m_{t-1} = K_t m_t + k_t, \qquad C_{t-1} = K_t C_t K_t^\top + \sigma_t^2 I_d, \qquad t = T, \dots, 2.$$

We run this recursion for all $T-1$ steps, and $p_1^{\mathbf{w}} = \mathcal{N}(m_1, C_1)$ is the exact law of the output of the discretized sampler. It is compared with $q_1^i = \mathcal{N}(\sqrt{\bar\alpha_1}\mu_i, \Sigma_1^i)$, as in (5).

Gram matrices. The score difference is affine, $D_{ij}(x,t) = B_{ij}^t(x - \sqrt{\bar\alpha_t}\mu_i) + c_{ij}^t$ with $B_{ij}^t := P_t^i - P_t^j$ and $c_{ij}^t := \sqrt{\bar\alpha_t}\, P_t^j(\mu_j - \mu_i)$, so the entries of (7) are

$$\big[\Sigma_i^{(t)}\big]_{jk} = c_{ij}^t \cdot c_{ik}^t + \operatorname{tr}\big(B_{ij}^t\, \Sigma_t^i\, (B_{ik}^t)^\top\big).$$

No estimation is involved. For a common covariance, $B_{ij}^t = 0$, and the expression reduces to the one in the proof of Corollary 3(b).

Total variation. For two Gaussians p, q on $\mathbb{R}^2$, we use $\mathrm{TV}(p,q) = \int (p-q)_+ = \int (q-p)_+$. Each integral is computed with the trapezoidal rule on a 341×341 grid covering $[-8.5, 8.5]^2$ in the coordinates that whiten the first Gaussian of the integrand, and the two values are averaged. On pairs with a common covariance, where the total variation has the closed form used in Appendix C, the error is below 10^{-5}, and on pairs taken from the experiments it is at most 4×10^{-5} when compared with a much finer quadrature. The threshold 10^{-4} used to decide that one weight beats another is above this error.

F.3. Computation of the weights

Minimax weights. The schedule is computed from the cone program of Proposition 4(a), with $(T-1)m + 1$ variables, written with CVXPY and solved with the interior-point solver Clarabel at its default tolerances. We pass the constraints $V_i(\mathbf{w}) \le \tau$ as sums of squares, one small cone

per term, a formulation the solver handles more reliably than m cones of dimension of order T. If the solver does not report an optimal status, we also solve the single-cone formulation and keep the schedule with the smaller value of $\max_i V_i$. The constant weight solves the m-variable program $\min_{w\in\Delta^m} \max_i w^\top G_i w$. For $m = 2$, both agree with the explicit solutions of Proposition 5 to a relative accuracy of 10^{-6} in the value, and for $m = 3$ and $m = 5$, we verified on a sample of instances that the value of the returned schedule is within a relative 2×10^{-5} of the dual bound of Proposition 4(b). Steps with a very small $1-\alpha_t$ contribute almost nothing to V_i, so the corresponding $w_t^\star$ are not well determined, and they also have almost no effect on the sampler.

Uniform-average rule. The same m-variable program with G_i replaced by $\frac{1}{T}\sum_{t=1}^{T} \Sigma_i^{(t)}$.

Random weights. Uniform on Δ^m, that is, Dirichlet$(1, \dots, 1)$. The rule that redraws at every step draws $w_2, \dots, w_T$ independently from this distribution.

Best constant weight found. The search starts from the best of the 103 constant weights evaluated earlier on the instance (the minimax constant weight, equal weights, the uniform-average weight, and the 100 random draws). It then performs seven rounds. In each round, it evaluates F_T at 64 Gaussian perturbations of the current best weight, with standard deviation 0.15, 0.07, 0.03, 0.015, 0.007, 0.003, and 0.0015 per coordinate, clipped to be nonnegative and renormalized, and keeps the best. This costs 448 further evaluations of F_T. The result is an upper bound on $\min_{w\in\Delta^m} F_T(w)$. We did not search over schedules.

F.4. Seeds and statistics

Seeds. Every configuration is run with seeds $0, \dots, 99$. For a configuration and a seed, we create a NumPy `SeedSequence` from the pair (CRC-32 checksum of the configuration name, seed) and spawn four independent generators from it, used respectively for the targets, the fixed random weights, the per-step random schedules, and the direct search. Changing the number of random draws therefore does not change the targets, and adding a configuration does not change the others. For the fixed instance, the first generator is unused.

Sample sizes. Each configuration has 100 instances, and each instance has 100 fixed random weights and 20 per-step random schedules. In total, 10,000 fixed random draws and 2,000 per-step random schedules per configuration.

Statistics. Because the instances are random, the worst-case gap varies much more between instances than between rules: its standard deviation over instances is between 0.09 and 0.24, depending on the rule and the configuration. All comparisons are therefore paired.

Since the computation is exact, the only variability is across instances and across random draws. Table 1 reports means over instances. Table 3 reports, for every other rule, the difference between its worst-case gap and that of the minimax schedule on the same instance (median and interquartile range over the 100 instances), and the share of instances on which the other rule beats ours by more than 10^{-4}. For the two random rules, the per-instance value is the mean over the draws, and the last column is the share of individual draws that beat ours. We do not report significance tests. With 100 paired instances and effects of this size, they carry no information beyond the sign counts in the last column.

Covariances	m	Compared with	Excess worst-case TV, median [IQR]	Beats ours
Common	2	Random weights (fixed)	$+0.151$ [$+0.115$, $+0.175$]	0%
		Random weights (redrawn every step)	$+0.017$ [$+0.012$, $+0.021$]	0%
		Equal weights	0.000 [0.000, 0.000]	0%
		Uniform-average rule	0.000 [0.000, 0.000]	0%
		Best constant weight found	0.000 [0.000, 0.000]	0%
Common	3	Random weights (fixed)	$+0.142$ [$+0.124$, $+0.152$]	0%
		Random weights (redrawn every step)	$+0.062$ [$+0.036$, $+0.082$]	0%
		Equal weights	$+0.062$ [$+0.037$, $+0.084$]	0%
		Uniform-average rule	0.000 [0.000, $+0.055$]	0%
		Best constant weight found	0.000 [0.000, 0.000]	7%
Common	5	Random weights (fixed)	$+0.104$ [$+0.089$, $+0.119$]	0%
		Random weights (redrawn every step)	$+0.060$ [$+0.038$, $+0.091$]	0%
		Equal weights	$+0.060$ [$+0.038$, $+0.089$]	0%
		Uniform-average rule	$+0.017$ [0.000, $+0.060$]	0%
		Best constant weight found	0.000 [0.000, 0.000]	8%
Different	2	Random weights (fixed)	$+0.133$ [$+0.111$, $+0.150$]	6%
		Random weights (redrawn every step)	$+0.011$ [$+0.004$, $+0.018$]	31%
		Equal weights	$+0.004$ [-0.001, $+0.014$]	33%
		Uniform-average rule	$+0.040$ [$+0.013$, $+0.070$]	7%
		Best constant weight found	-0.019 [-0.031, -0.009]	98%
Different	3	Random weights (fixed)	$+0.115$ [$+0.085$, $+0.141$]	5%
		Random weights (redrawn every step)	$+0.035$ [$+0.013$, $+0.077$]	16%
		Equal weights	$+0.034$ [$+0.011$, $+0.077$]	14%
		Uniform-average rule	$+0.044$ [$+0.015$, $+0.073$]	11%
		Best constant weight found	-0.024 [-0.039, -0.009]	93%
Different	5	Random weights (fixed)	$+0.092$ [$+0.065$, $+0.118$]	4%
		Random weights (redrawn every step)	$+0.049$ [$+0.023$, $+0.072$]	10%
		Equal weights	$+0.049$ [$+0.021$, $+0.073$]	11%
		Uniform-average rule	$+0.039$ [$+0.012$, $+0.068$]	16%
		Best constant weight found	-0.024 [-0.039, -0.011]	96%

Table 3: Paired comparison with the minimax schedule over 100 random instances: worst-case TV of the other rule minus ours on the same instance, as median and interquartile range (IQR). Positive values mean that ours is better. For the random rules, the per-instance value is the mean over the draws, and *Beats ours* is the share of individual draws with a smaller worst-case TV than ours. For the other rules, it is the share of instances.

F.5. Paired results

Table 3 and Figure 1 give the paired version of Table 1. Three observations complement Section 4.4. First, the rule that redraws a random weight at every step is indistinguishable from equal weights for $m = 3$ and $m = 5$ (medians $+0.062$ against $+0.062$, $+0.060$ against $+0.060$, $+0.035$ against $+0.034$, $+0.049$ against $+0.049$). Over $T - 1$ steps, the independent draws average out, and the effective weight is close to the mean weight $1/m$. It is therefore not an independent baseline, so Table 1 omits it. Second, for a common covariance, the uniform-average rule coincides with the minimax weights on at least half of the instances with $m = 3$ (median 0.000) but is much worse

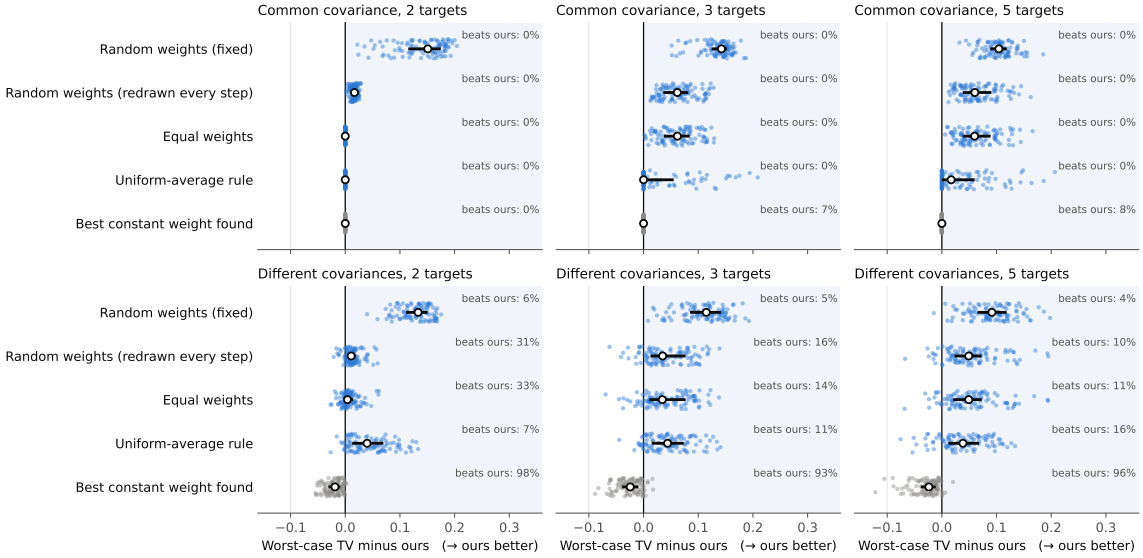


Figure 1: Each panel is one configuration and each row one competing rule. A dot is one random instance (100 per row), and its position is the worst-case total variation of that rule minus that of the minimax schedule on the same instance, so dots to the right of the vertical line (shaded region) are instances on which our weights are better. For the two random rules, a dot is the mean over the draws of that instance. The white marker and the black bar are the median and the interquartile range. *Beats ours* is the share of instances (for the random rules, of individual draws) on which the rule has a smaller worst-case gap than ours. The last row is the reference found by direct search, not a practical rule.

on others (upper quartile $+0.055$). The two rules agree when the active targets are the same in both metrics and disagree when the uniform average assigns weight zero to a target. Third, for different covariances, the negative values in the last row of each panel quantify the suboptimality of the minimax schedule, and the largest observed loss with respect to the best constant weight found is 0.054, 0.089, and 0.122 for $m = 2, 3, 5$.

F.6. Computing environment and running time

Hardware. All experiments were run on a single laptop, an Apple MacBook Pro (model identifier Mac16,6) with an Apple M4 Max processor (14 cores, 10 performance and 4 efficiency), 36 GB of memory, and macOS 27.0.1. No GPU is used.

Software. Python 3.11.17, NumPy 2.4.6, SciPy 1.17.1, CVXPY 1.9.3, and Clarabel 0.11.1. Seeds are distributed over 12 worker processes, each restricted to a single linear-algebra thread.

Running time. Table 4 lists the wall-clock time of each configuration, and the whole experiment takes 44 minutes. For one instance with $T = 1000$, solving the cone program for the minimax schedule takes between 0.1 and 1 second depending on m, and the constant-weight program is faster still. The time is dominated by the evaluations of F_T, each of which runs the $T - 1$ steps

Configuration	Covariances	m	T	Wall-clock time
Fixed instance	Common	3	3000	319 s
Random instances	Common	2	1000	223 s
Random instances	Common	3	1000	343 s
Random instances	Common	5	1000	569 s
Random instances	Different	2	1000	228 s
Random instances	Different	3	1000	345 s
Random instances	Different	5	1000	589 s
Total				2616 s (44 min)

Table 4: Wall-clock time of each configuration (100 seeds, 12 worker processes) on the machine described in the text.

of the moment recursion and computes $2m$ two-dimensional integrals. Every instance requires 103 evaluations for the constant weights, 21 for the schedules, and 448 for the direct search. Without the direct search, the whole experiment takes about 10 minutes.

Code. The script `experiments/run_random_weights.py` runs the experiment and writes all per-seed values to `experiments/results/random_weights.json`. The script `experiments/plot_random_weights.py` produces the tables and Figure 1 from that file, and `experiments/sanity_checks.py` checks the closed forms, the quadrature, and the solvers against independent computations.